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Estimation of Lateral Velocity and Cornering Stiffness in Vehicle Dynamics Based on Multi-Source Information Fusion

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Abstract

To address the challenge of directly measuring essential dynamic parameters of vehicles, this article introduces a multi-source information fusion estimation method. Using the intelligent front camera (IFC) sensor to analyze lane line polynomial information and a kinematic model, the vehicle's lateral velocity and sideslip angle can be determined without extra sensor expenses. After evaluating the strengths and weaknesses of the two aforementioned lateral velocity estimation techniques, a fusion estimation approach for lateral velocity is proposed. This approach extracts the vehicle's lateral dynamic characteristics to calculate the fusion allocation coefficient. Subsequently, the outcomes from the two lateral velocity estimation techniques are merged, ensuring rapid convergence under steady-state conditions and precise tracking in dynamic scenarios. In addition, we introduce a tire parameter online adaptive module (TPOAM) to continually update essential tire parameters such as cornering stiffnesses, with its effectiveness demonstrated through DLC and slalom simulation tests. Using a dual extended Kalman filter (DEKF) observer, the article allows for joint estimation of vehicle states and tire parameters. Ultimately, we offer a cost-effective estimation method of vital dynamic vehicle parameters to support the motion control module in autonomous driving.

History

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Introduction

Critical dynamic parameters of vehicles, including sideslip angle, yaw rate, longitudinal speed, and cornering stiffness, are vital for the stability and motion control systems in autonomous driving [1, 2, 3]. The inertial measurement unit (IMU) sensor can directly measure the yaw rate, while the longitudinal speed can be derived from the rotational speed of the non-driven wheels. The sideslip angle, which characterizes the vehicle's stability, is crucial for ensuring precise motion tracking and optimizing the vehicle's stability control system [4, 5, 6]. However, it cannot be directly measured from inexpensive sensors [7, 8, 9]. The current methods for estimating the vehicle's sideslip angle are as follows:

I. ***Estimation method based on GPS/INS*** [10, 11, 12, 13]: GPS provides low-frequency data for position and heading, while the IMU offers high-frequency measurements of the vehicle's state. By integrating the vehicle's low-frequency position and heading data with its motion state, the vehicle's lateral velocity and sideslip angle can be determined [12]. Wei Liu et al. proposed a novel kinematic-model-based vehicle slip angle estimation method by fusing information from a global navigation satellite system (GNSS) and an IMU. To negate the gravitational components influenced by the vehicle's roll and pitch, a vehicle attitude angle observer that utilizes the square-root cubature Kalman filter (SCKF) is employed, aiming to provide precise estimates of roll and pitch [13]. The practice of estimating sideslip angles using a GPS/INS combination is already well-established. Furthermore, specialized equipment such as GPS/IMU has been incorporated into certain autonomous vehicles. High-precision vehicle state measurement and calibration equipment such as the RT3000 V3 can yield accurate measurements of the sideslip angle and lateral velocity. However, due to financial constraints, these advanced methods are not yet widespread in mass-produced vehicles. Additionally, certain environments, like areas with tall buildings or tunnels, impose limitations on these techniques.

II. ***Estimation method based on the kinematic model*** [14, 15, 16]: Based on the formula used to calculate the vehicle sideslip angle, one can determine the rate of change of the sideslip angle. By integrating this rate, the magnitude of the sideslip angle can be estimated. Liang Li et al. proposed an innovative variable structure extended Kalman filter (VSEKF) for the vehicle sideslip angle estimation based on the kinematic model. This VSEKF incorporates a feedback component related to the rate of the sideslip angle, aiding in compensating for errors introduced by road friction and in dampening the accumulation of errors due to noise in the real sideslip angle rate [14]. The kinematic model-based approach to estimating sideslip angle proves effective, especially in transient scenarios where the sideslip angle changes rapidly. However, challenges arise when using this method. Integrative calculations, combined with the interference of measurement noise from the IMU sensor, can lead to significant drift in sideslip angle observations in steady-state conditions. This drift often results from the compounding effects of integration errors [15]. Therefore, it's concluded that the kinematic model-based method be combined with other estimation techniques for optimal results [16].

III. ***Estimation method based on the dynamic model*** [4, 17, 18, 19, 20]: By utilizing vehicle dynamics data such as yaw rate, longitudinal and lateral accelerations, specific dynamics, and tire models are chosen to design observers to estimate the vehicle's lateral velocity. Salvatore Strano et al. focused on vehicle sideslip angle estimation, adopting a constrained unscented Kalman filter (CUKF) that takes into account state constraints during the estimation process. State boundaries are useful in real-world applications to prevent unphysical results and to improve the estimator's robustness. The approach gives ample consideration to measurement noise and nonlinearities. Moreover, a single-track vehicle model is used in the design of the estimator [18]. However, it's worth noting that the efficacy of this estimation technique, especially for the sideslip angle or lateral velocity, is contingent on the precision of the model parameters [17, 19].

IV. ***Estimation method based on multi-source information fusion*** [16, 21, 22, 23, 24]: Due to the limitations of individual methods for determining sideslip angles or lateral velocities, the academic community is showing a rising interest in methods that fuse information from multiple sources. Li Xiaoyu et al. proposed a fusion estimation method based on the kinematic method and kinematic–geometry method. Their approach employs an extended Kalman filter (EKF) grounded in a 3-DoF vehicle kinematics model. Alongside this, they've incorporated a method rooted in the kinematic–geometry relationship, enabling the estimation of both dynamic and steady-state values of the vehicle's sideslip angle. A specialized data fusion technique, which centers on the dynamic properties of the vehicle sideslip angle, has been engineered to enhance the precision of the concluding estimation [16].

V. ***Estimation method based on neural network*** [25, 26, 27, 28, 29]: Traditional model-based methods for sideslip angle estimation rely heavily on accurate vehicle parameters. In contrast, NN-based methods can operate without these exact parameters, provided there's enough training data. The potential of NN-based sideslip angle estimation has been evident, especially in the realm of vehicle parameter estimation. The efficacy of this method largely depends on the volume and quality of the training data available. Torben Gräber et al. proposed a supervised machine learning state estimation scheme that is able to estimate the current vehicle sideslip angle. It consists of a recurrent neural network (RNN)

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with gated recurrent units (GRUs), an additional input projection, and a regression head. The architecture was specifically chosen to strike a balance: it aims to reduce the computational demands of the model without sacrificing its capability to capture and express complex patterns and behaviors [25].

The cornering stiffnesses of both the front and rear axles are pivotal parameters in the single-track dynamic model, which denotes the proportional relationship between the sideslip angle and the tire's lateral force. Within the vehicle's dynamic model, the combined cornering stiffness of the front and rear axles equates to the sum of the cornering stiffness of both the left and right wheels on these axles. The accuracy of these values has a direct bearing on the precision of the lateral response in the vehicle's dynamic modeling. Currently, two prevalent methods to determine the cornering stiffnesses are:

I. ***Calibration method via tire test bench***: This technique employs a tire test bench to delineate the correlation between the tire's lateral force and the sideslip angle. This relationship is subsequently transformed into either a handy lookup table or an empirical tire formula suitable for online applications. In this realm, Xu Nan and his team's work stands out. Their comprehensive analysis of extensive data from rubber friction and high-speed tire test benches revealed that the variability in tire coefficients can be mathematically modeled using both uniform and normal distribution patterns. In response to this, they developed the UniTire stochastic model. By leveraging the general polynomial chaos (gPC) methodology, this model adeptly captures the tire's mechanical behavior in the face of uncertainties, offering a more precise depiction of the tire's mechanical attributes [30].
II. ***Model-based estimation method***: This approach employs a vehicle dynamics model to determine cornering stiffness using the vehicle's response to steering angle and longitudinal velocity inputs. As measurements or estimates of the sideslip angle have become more accessible, a growing number of researchers are adopting this method to ascertain cornering stiffness. Techniques include the GPS/INS-based method [10, 31], observer-based method [9, 32, 33], and least squares-based method [34, 35]. The ***GPS/INS-based method*** utilizes GPS/INS of lateral velocity and yaw rate to derive cornering stiffness. However, this method may produce significant fluctuations in the estimated values for front and rear axle cornering stiffness, primarily due to environmental noise during real-world vehicle operation. The ***observer-based method*** centers on the design of multiple observers that collaboratively estimate both the state and cornering stiffness of the vehicle. Currently, popular observers in this category encompass tools such as the Kalman filter, Luenberger observer, and sliding mode observer. Seungyong LEE et al. proposed a novel identification method using a dual Kalman filter algorithm and a GPS measurement system to estimate the cornering stiffness for on-board identification [36]. The ***least squares-based method*** harnesses the vehicle dynamics model and the vehicle's lateral response to align with the cornering stiffness of the front and rear axles via the least squares method. Kyoungseok Han and collaborators explored the interplay between road surface conditions and tire cornering stiffness. Recognizing that tire cornering stiffness exhibits considerable variability based on road conditions, it can be inferred from the tire's initial, accurate cornering stiffness measurements. They juxtaposed their calculated results with findings from the RT3000 estimation [34]. This approach necessitates leveraging historical vehicle states from buffers, leading to some inevitable estimation delays, making it more suitable for offline cornering stiffness calibration.

To sum up, the majority of techniques to estimate vital vehicle dynamics parameters lean heavily on dynamic or kinematic models. Frequently adopted methods often grapple with issues such as state drift during steady-state scenarios or substantial estimation inaccuracies stemming from model parameter discrepancies. While high-precision vehicle state measurement equipment can augment accuracy, it comes with increased sensor costs, making large-scale mass production less feasible in the short term. Roads with structured designs offer distinct characteristics, such as minimal curvature variance and change rates. Detected lane lines and roadside harbor a wealth of data related to vehicle movement. By analyzing data from the IFC sensor, direct measurements of vehicle deviations and motion states can be procured. This information can serve as a foundation for estimating key vehicle dynamics parameters. However, fusing data from multiple sources presents both immense opportunities and challenges for vehicle state estimation. It's worth noting that the current techniques for estimating essential vehicle dynamics parameters, especially those relying on multi-source information fusion, are still in their nascent stages. In this article, our contributions are summarized as follows:

I. This article introduces a method where lane line information, as detected by the IFC sensor, is used. By comparing outputs from two consecutive frames, the vehicle's local posture is deduced, which then allows for the calculation of its lateral velocity V_{y_IFC};
II. This article presents a fusion technique by utilizing the vehicle lateral dynamic characteristics. This technique integrates the lateral velocity derived from the IFC sensor V_{y_IFC} with $V_{y_Kinematic}$ estimated from kinematic models, thus providing a more comprehensive estimation of the vehicle's lateral velocity V_{y_Fusion};
III. The tire parameter online adaptive module (TPOAM) is introduced in this research. This module is designed to simultaneously estimate the vehicle's state and the equivalent cornering stiffnesses for both the front and rear axles.

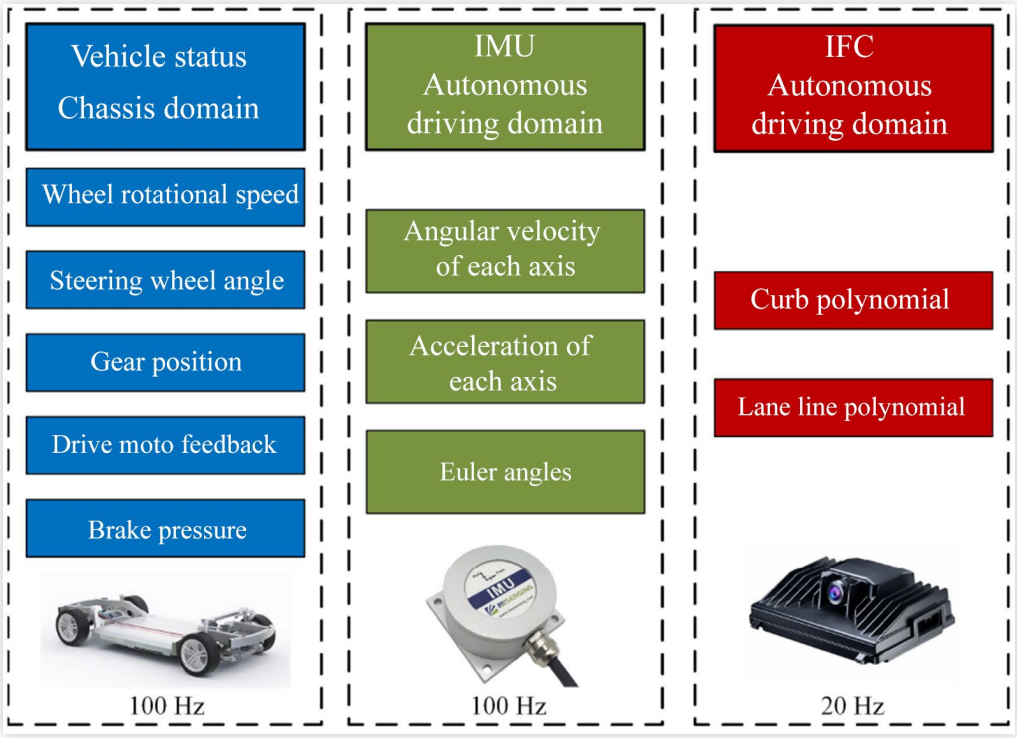
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Overall Framework of the Estimator of Key Parameters in Vehicle Dynamics Based on Multi-Source Information Fusion

Under the domain control architecture of the autonomous vehicle, the sensor information available in the chassis domain and the self-driving domain mainly includes vehicle status information fed back from the chassis domain, IFC sensor detection information in the self-driving domain, and vehicle motion status information detected by IMU, as shown in Figure 1. The feedback vehicle state through the chassis domain is output through controller area network (CAN), which includes wheel rotational speed, steering wheel angle, gear position, drive motor feedback, and brake pressure, with an update frequency of 100 Hz. The IMU measurement signals in the autonomous driving domain are output through CAN or serial ports, and the obtained signals include angular velocity of each axis, acceleration of each axis, and Euler angles, with an update frequency of 100 Hz. The road information obtained by the IFC sensor in the autonomous driving domain is mainly communicated through the CAN, including curb polynomial and lane line polynomial information, with an update frequency of 20 Hz.

The overall framework of the proposed estimator of key parameters in vehicle dynamics based on multi-source information fusion is composed of four parts, as shown in the Figure 2. The first module is the vehicle lateral velocity estimation module based on IFC sensor, which mainly utilizes the lane line polynomial information to estimate the lateral velocity of the vehicle $V_{y_}$IFC. The second module is the vehicle lateral velocity estimation module based on kinematic model, which uses the vehicle state output from the IMU to estimate the vehicle lateral velocity based on the kinematic model $V_{y_}$Kinematic. Notably, two estimation modules do not require vehicle dynamics parameters such as the cornering stiffnesses of the front and rear axles. The third module fuses the lateral velocity estimated values $V_{y_}$IFC and $V_{y_}$Kinematic by comparing the advantages and disadvantages of the above two lateral velocity estimation modules. The fusion estimation of the lateral velocity $V_{y_}$Fusion can be calculated by extracting the vehicle's lateral dynamic characteristics and calculating the fusion allocation coefficient. The fourth module is the TPOAM, which takes the lateral velocity fusion estimation value $V_{y_}$Fusion and the yaw rate r to jointly estimate the cornering stiffness of the front and rear axles as well as the vehicle state. The four modules mentioned above are described in detail next.

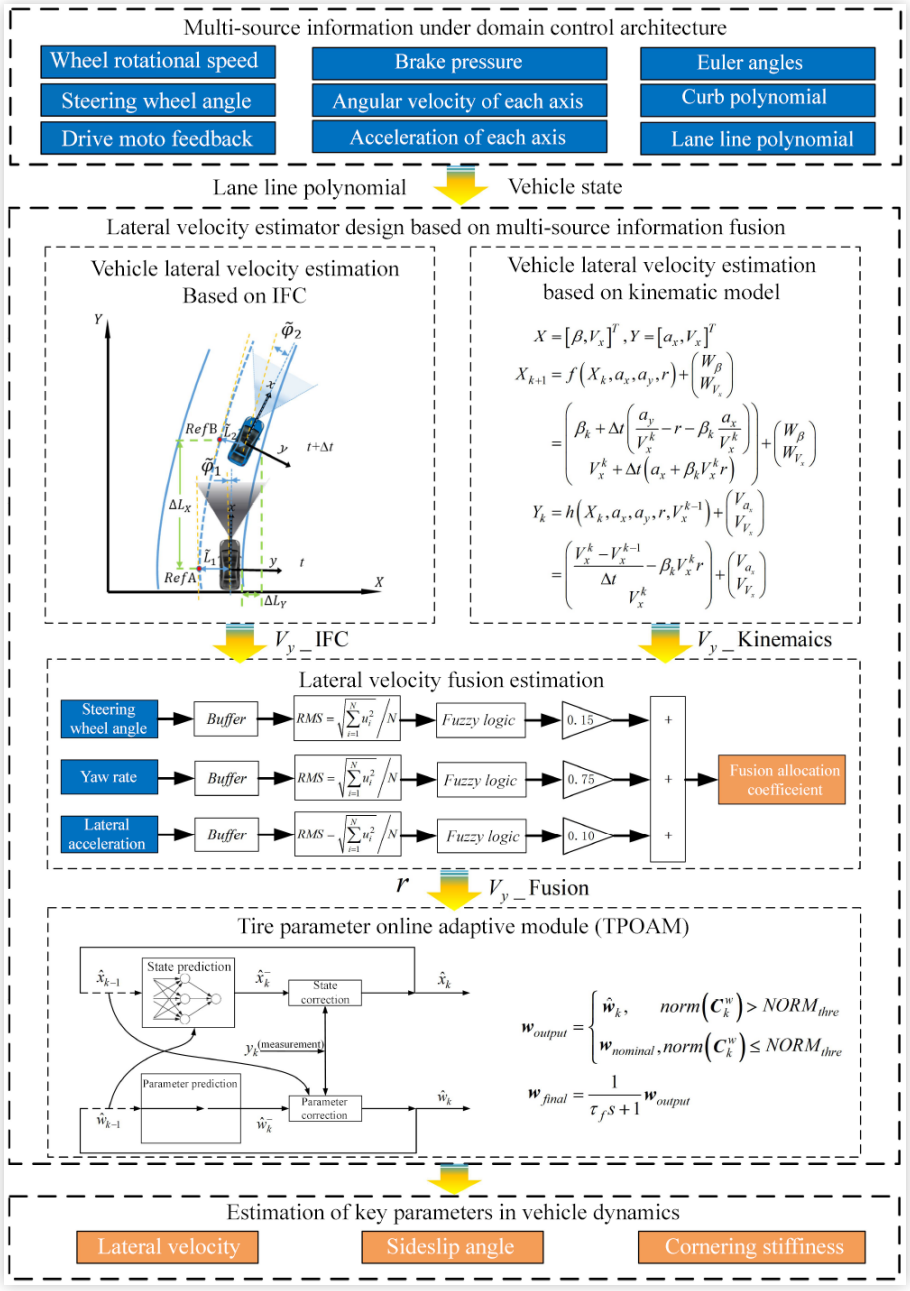
FIGURE 1 List of multi-source information.



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FIGURE 2 Overall framework for estimator of key parameters in vehicle dynamics based on multi-source information fusion.



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Vehicle Lateral Velocity Estimation Based on IFC Sensor

The vehicle lateral velocity estimation based on IFC sensor can be divided into four steps: prediction of vehicle state, calculation of the deviation between the vehicle and the reference point, estimation of vehicle lateral and longitudinal velocity, and filtering of estimated lateral velocity. The workflow of vehicle lateral velocity estimation based on IFC sensor is shown as follows.

I. *Prediction of vehicle state*: In this article, constant turn rate and velocity (CTRV) model is adopted to predict the current vehicle state [37]. The CTRV model has a great advantage in predicting the state of a target that is moving in a curved line on a plane, which assumes that the vehicle's longitudinal velocity and yaw rate are constant. The vehicle state at this moment is predicted using the vehicle state at the previous moment, and the predicted state at this moment is projected into a local coordinate with the centerline of the

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lane at the previous moment as the reference line. Select state as $X = [x, y, yaw]^T$, where x, y, and yaw are the horizontal coordinate, vertical coordinate, and the heading angle of the vehicle in the local coordinate system established at the last moment. At the last moment, the vehicle is located at the origin position, and the direction of the vehicle is the positive direction of the x-axis of the local coordinate system, which means:

$$X_{k-1} = \left[0,0,0\right]^T \quad \text{Eq. (1)}$$

The longitudinal velocity V_x^{k-1} and the yaw rate r_{k-1} of the vehicle at the previous moment are taken as the input, and the state of the vehicle at the current moment in the local coordinate established in the previous moment can be calculated according to the CTRV model:

$$X_k = \begin{bmatrix} x_{pred} \\ y_{pred} \\ yaw_{pred} \end{bmatrix} = X_{k-1} + \begin{bmatrix} \int_0^{T_{IFC}} V_x^{k-1} \cos\left(r_{k-1} t\right) dt \\ \int_0^{T_{IFC}} V_x^{k-1} \sin\left(r_{k-1} t\right) dt \\ r_{k-1} T_{IFC} \end{bmatrix} = \begin{bmatrix} \int_0^{T_{IFC}} V_x^{k-1} \cos\left(r_{k-1} t\right) dt \\ \int_0^{T_{IFC}} V_x^{k-1} \sin\left(r_{k-1} t\right) dt \\ r_{k-1} T_{IFC} \end{bmatrix} \quad \text{Eq. (2)}$$

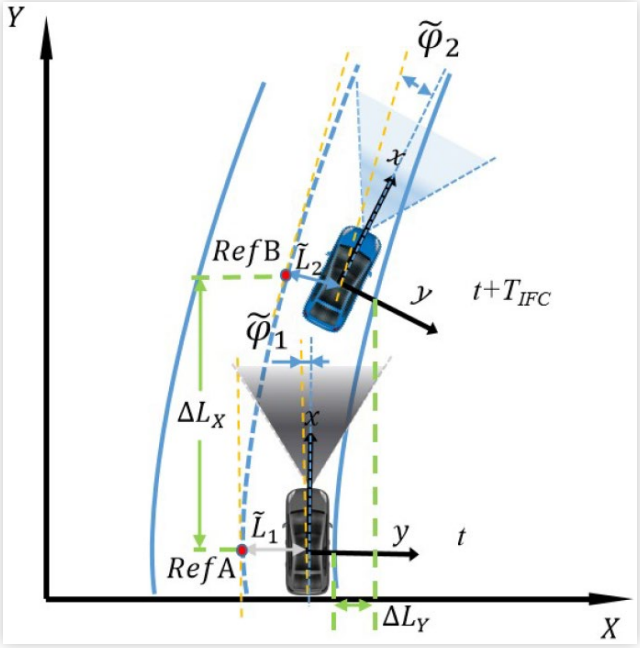
where T_{IFC} is the update cycle of IFC sensor.

II. *Calculation of the deviation between the vehicle and the reference point*: Schematic of vehicle lateral velocity estimation based on IFC sensor is shown as Figure 3. The lateral deviation and heading deviation between the vehicle and the centerline of the lane line at the moments t and $t + \Delta t$ can be obtained from the IFC sensor: $\tilde{L}_1, \tilde{L}_2$ and $\tilde{\varphi}_1, \tilde{\varphi}_2$. *Ref* A is the lane centerline reference point of this vehicle at the previous moment, *Ref* B is the lane centerline reference point of this vehicle at the previous moment.

Based on the predicted state of the vehicle at this moment x_k and the polynomial coefficients of the centerline of the lane line at the previous moment $\left[A_0^{k-1}, A_1^{k-1}, A_2^{k-1}, A_3^{k-1}\right]$, the lateral deviation L_{pred} and heading deviation φ_{pred} of the *Ref* B respect to the local coordinate system established at the previous moment can be calculated:

$$\begin{aligned} L_{pred} &= A_3^{k-1} x_{pred}^3 + A_2^{k-1} x_{pred}^2 + A_1^{k-1} x_{pred} + A_0^{k-1} \\ \varphi_{pred} &= \tan^{-1}\left(3A_3^{k-1} x_{pred}^2 + 2A_2^{k-1} x_{pred} + A_1^{k-1}\right) \end{aligned} \quad \text{Eq. (3)}$$

FIGURE 3 Schematic of vehicle lateral velocity estimation based on IFC sensor.



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At the same time, the longitudinal, lateral, and heading deviations ΔL_X, ΔL_Y, $\Delta\varphi_{lane}$ of the *Ref* B relative to the *Ref* A can be calculated as follows:

$$\begin{aligned} \Delta L_X &= x_{pred} \\ \Delta L_Y &= L_{pred} - \tilde{L}_1 \\ \Delta\varphi_{lane} &= \varphi_{pred} - \tilde{\varphi}_1 \end{aligned} \quad \text{Eq. (4)}$$

III. *Estimation of vehicle lateral and longitudinal velocity*: Based on the heading deviation $\tilde{\varphi}_1$ of the *Ref* A relative to the vehicle at moment t, the heading deviation $\tilde{\varphi}_2$ of the *Ref* B relative to the vehicle at moment $t + \Delta t$, and the heading deviation $\Delta\varphi_{lane}$ of the *Ref* B relative to the *Ref* A, the rotation angle $\hat{\theta}$ of the vehicle at the moment $t + \Delta t$ relative to the vehicle at the moment t can be calculated:

$$\hat{\theta} = -\tilde{\varphi}_2 + \Delta\varphi_{lane} + \tilde{\varphi}_1 = -\tilde{\varphi}_2 + \varphi_{pred} \quad \text{Eq. (5)}$$

According to the longitudinal and lateral deviations ΔL_X, ΔL_Y of the *Ref* B relative to the *Ref* A, and the lateral deviation $\tilde{L}_1, \tilde{L}_2$ of the reference point relative to the vehicle at the moment t and $t + \Delta t$, the projection $\hat{V}_X, \hat{V}_Y$ of the vehicle velocity at the moment $t + \Delta t$ to the local coordinate system established at the previous moment can be calculated:

$$\begin{aligned} \Delta X &= \Delta L_X \\ \Delta Y &= -\tilde{L}_2 \cos\Delta\varphi_{road} + \Delta L_Y + \tilde{L}_1 = -\tilde{L}_2 \cos\Delta\varphi_{road} + L_{pred} \\ \hat{V}_X &\approx \frac{\Delta X}{T_{IFC}} = \frac{\Delta L_X}{T_{IFC}} \\ \hat{V}_Y &\approx \frac{\Delta Y}{T_{IFC}} = \frac{-\tilde{L}_2 \cos\Delta\varphi_{road} + L_{pred}}{T_{IFC}} \end{aligned} \quad \text{Eq. (6)}$$

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Based on the above, the estimated vehicle lateral velocity V_y_IFC at the moment $t + \Delta t$ can be calculated through by rotating $\hat{\theta}$:

$$V_y_\text{IFC} = \hat{V}_Y \cos\hat{\theta} - \hat{V}_X \sin\hat{\theta} \quad \text{Eq. (7)}$$

IV. *Filtering of estimated lateral velocity*: In order to prevent the lateral velocity fluctuation caused by the lane centerline changing during the lane-changing process of this vehicle, this article designs a Kalman filter to filter the estimated lateral velocity V_y_IFC. The change rate of the vehicle lateral velocity $\dot{V}_y$ can be calculated as follows:

$$\dot{V}_y = a_y - V_x r \quad \text{Eq. (8)}$$

where a_y is the vehicle lateral acceleration. The state space equations for the Kalman filter for the lateral velocity V_y_IFC are as follows:

$$\begin{aligned} X &= \left[V_y, \dot{V}_y\right]^T, \quad Y = \left[V_y, \dot{V}_y\right]^T \\ X_{k+1} &= A_1 X_k + W_1 = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} X_k + \begin{pmatrix} W_{V_y} \\ W_{\dot{V}_y} \end{pmatrix} \\ Y_k &= C_1 X_k + V_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} X_k + \begin{pmatrix} V_{V_y} \\ V_{\dot{V}_y} \end{pmatrix} \end{aligned} \quad \text{Eq. (9)}$$

where $W_1 = \left[W_{V_y}, W_{\dot{V}_y}\right]^T$ is the process noise, $V_1 = \left[V_{V_y}, V_{\dot{V}_y}\right]^T$ is the measurement noise, W_1, V_1 are regarded as white noise with covariance Q_1, R_1, respectively. The process of the Kalman filter can be divided into two parts:

State prediction:

$$\begin{aligned} X_k^- &= A_1 X_{k-1} \\ P_{X_k}^- &= A_1 P_{X_{k-1}} A_1^T + Q_1 \end{aligned} \quad \text{Eq. (10)}$$

State correction:

$$\begin{aligned} K_k^X &= P_{X_k}^- C_1^T \left(C_1 P_{X_k}^- C_1^T + R_1\right)^{-1} \\ X_k &= X_k^- + K_k^X \left(Y_k - C_1 X_k^-\right) \\ P_{X_k} &= \left(I - K_k^X C_1\right) P_{X_k}^- \end{aligned} \quad \text{Eq. (11)}$$

Vehicle Lateral Velocity Estimation Based on Kinematic Model

The vehicle lateral velocity estimation based on the kinematic model utilizes the integration of $\dot{\beta}$ to obtain the sideslip angle of the vehicle. According to the formula for calculating the sideslip angle: $\beta = \frac{V_y}{V_x}$, $\dot{\beta}$ can be derived:

$$\dot{\beta} = \frac{\dot{V}_y V_x - V_y \dot{V}_x}{V_x^2} = \frac{\dot{V}_y}{V_x} - \beta \frac{\dot{V}_x}{V_x} \quad \text{Eq. (12)}$$

Due to the small size of β, ignoring the impact of including the β^2 term on $\dot{\beta}$ yields:

$$\begin{aligned} a_x &= \dot{V}_y + V_x r \\ a_y &= \dot{V}_x - V_y r \\ \dot{\beta} &= \frac{a_y}{V_x} - r - \beta \frac{a_x}{V_x} \end{aligned} \quad \text{Eq. (13)}$$

where a_x is the vehicle longitudinal acceleration. Based on kinematic model, the sideslip angle β can be obtained by integrating the above equation:

$$\beta = \int \left(\frac{a_y}{V_x} - r - \beta \frac{a_x}{V_x} \right) dt \quad \text{Eq. (14)}$$

The estimated vehicle lateral velocity based on the kinematic model V_y_Kinematic can be calculated by combining with the vehicle longitudinal velocity which can be calculated from the chassis feedback information of the non-driven wheel rotational speed. The vehicle yaw rate, longitudinal and lateral acceleration can be obtained by the IMU sensor. The integral term β is inputted by the estimation value of the previous moment. The state space equation for estimating vehicle lateral velocity based on the kinematic model is as follows:

$$\begin{aligned} X &= \left[\beta, V_x\right]^T, \quad Y = \left[a_x, V_x\right]^T \\ X_{k+1} &= f\left(X_k, a_x, a_y, r\right) + \begin{pmatrix} W_\beta \\ W_{V_x} \end{pmatrix} \\ &= \begin{pmatrix} \beta_k + T_{IMU}\left(\frac{a_y}{V_x^k} - r - \beta_k \frac{a_x}{V_x^k}\right) \\ V_x^k + T_{IMU}\left(a_x + \beta_k V_x^k r\right) \end{pmatrix} + \begin{bmatrix} W_\beta \\ W_{V_x} \end{bmatrix} \\ Y_k &= h\left(X_k, a_x, a_y, r, V_x^{k-1}\right) + \begin{pmatrix} V_{a_x} \\ V_{V_x} \end{pmatrix} \\ &= \begin{pmatrix} \frac{V_x^k - V_x^{k-1}}{T_{IMU}} - \beta_k V_x^k r \\ V_x^k \end{pmatrix} + \begin{pmatrix} V_{a_x} \\ V_{V_x} \end{pmatrix} \end{aligned} \quad \text{Eq. (15)}$$

where T_{IMU} is update cycle of IMU sensor, $W_2 = [W_\beta, W_{V_x}]^T$ is the process noise, $V_2 = [V_{a_x}, V_{V_x}]^T$ is the measurement noise, W_2, V_2 are regarded as white noise with covariance Q_2, R_2, respectively.

Due to the nonlinearity involved in the state space equation and measurement equation, it is necessary to use an EKF to estimate the vehicle sideslip angle [38]. The Jacobian matrix required in the EKF can be calculated as follows:

$$\begin{aligned} A_2 &= \frac{\partial f\left(X_k, a_x, a_y, r\right)}{\partial X_k} = \begin{bmatrix} 1 - T_{IMU}\frac{a_x}{V_x^k} & T_{IMU}\frac{\beta_k a_x - a_y}{\left(V_x^k\right)^2} \\ T_{IMU} V_x^k r & 1 + T_{IMU}\beta_k r \end{bmatrix} \\ C_2 &= \frac{\partial h\left(X_k, a_x, a_y, r, V_x^{k-1}\right)}{\partial X_k} = \begin{bmatrix} -V_x^k r & \frac{1}{T_{IMU}} - \beta_k r \\ 0 & 1 \end{bmatrix} \end{aligned} \quad \text{Eq. (16)}$$

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The specific process of the EKF can also be divided into two parts:

State prediction:

$$X_k^- = f\left(X_{k-1}, a_x, a_y, r\right)$$
$$P_{X_k}^- = A_2 P_{X_{k-1}} A_2^T + Q_2 \quad \text{Eq. (17)}$$

State correction:

$$K_k^X = P_{X_k}^- C_2^T \left(C_2 P_{X_k}^- C_2^T + R_2\right)^{-1}$$
$$X_k = X_k^- + K_k^X \left(Y_k - h\left(X_k^-, a_x, a_y, r, V_x^{k-1}\right)\right) \quad \text{Eq. (18)}$$
$$P_{X_k} = \left(I - K_k^X C_2\right) P_{X_k}^-$$

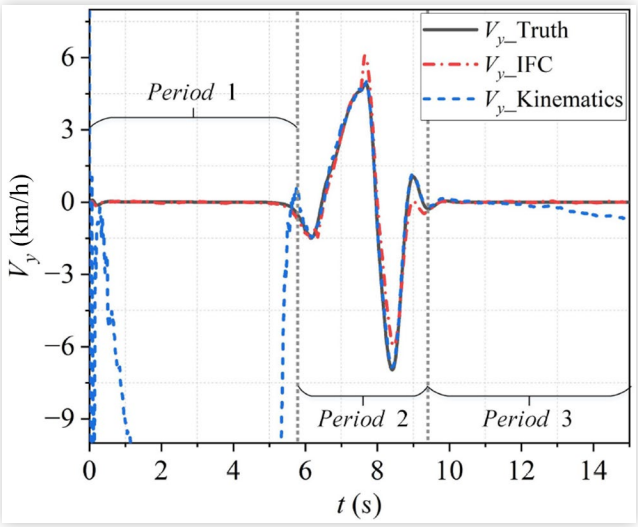
After estimating the sideslip angle $\beta_{kinematics}$ by the EKF, the vehicle lateral velocity based on the kinematic model V_y_Kinematic can be calculated as follows:

$$V_y_\text{Kinematic} = \beta_{kinematics} V_x \quad \text{Eq. (19)}$$

Lateral Velocity Fusion Estimation

The results of vehicle lateral velocity estimation based on IFC sensor and the kinematic model in the simulation environment are shown in Figure 4. From the lateral velocity estimation results, it can be seen that: (I) In the case where the vehicle lateral velocity tends to be steady state, for example, in Period 1 (0–5.8 s) and Period 3 (9.4–15 s), due to the influence of the sensor noise there will be evident drift and jitter. The lateral velocity estimation results of the vehicle based on the kinematics model have an obvious deviation from the truth value and cannot converge; (II) When the vehicle lateral velocity changes rapidly, such as Period 2 (5.8–9.4 s), the lateral velocity calculated by the vehicle lateral estimation algorithm based on the kinematic model can quickly and accurately converge to the truth value and has higher estimation accuracy compared to the vehicle lateral velocity estimation method based on IFC sensor; (III) The vehicle lateral speed estimation method based on IFC sensor does not cause any drift in the steady-state condition and can realize the accurate estimation of vehicle lateral velocity.

FIGURE 4 The results of vehicle lateral velocity estimation based on IFC sensor and the kinematic model in the simulation environment.



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In order to integrate the advantages of the above two lateral velocity estimation methods, the design of the vehicle lateral velocity fusion estimation algorithm is carried out next. Referring to the literature [21], the vehicle's front wheel angle of rotation, lateral acceleration, and yaw rate are utilized for the extraction of lateral dynamic characteristics, and the fusion allocation coefficient *Ratio* is calculated based on the lateral dynamic characteristics. Among these vehicle states, the variations of the lateral acceleration and the yaw rate are in good agreement with the change in the vehicle's lateral velocity. The vehicle's front wheel angle is obtained from the driver input or calculated by the autonomous motion controller. Compared with the rest of the states, the vehicle's front wheel angle can show the vehicle's lateral dynamic characteristics in advance.

To prevent the influence of sensor noise on the extraction of vehicle lateral dynamic characteristics and the calculation of fusion allocation coefficients *Ratio*, this article first buffers the mentioned three vehicle states required for dynamic extraction. The buffer length is determined by the quality of the states with sensor output (0.1 s is chosen in this article; too large a buffer will lead to a lag in the extraction of vehicle lateral dynamic characteristics). Root mean square error (RMSE) of the buffered state is calculated. The numerical representation of the vehicle lateral dynamic characteristics is carried out by using fuzzy logic. In this article, the trapezoidal function is chosen for the design of fuzzy rules. The trapezoidal function is calibrated to realize the extraction of the dynamic characteristics of vehicle lateral dynamics by comparing the changes of vehicle lateral speed, vehicle front wheel angle, lateral acceleration, and yaw rate under steady-state and transient conditions. The value of the fusion allocation coefficients *Ratio* can be obtained by weighted averaging the three fuzzy logic outputs as shown in Figure 5. The fusion estimation of the vehicle lateral velocity V_y_Fusion is calculated as follows:

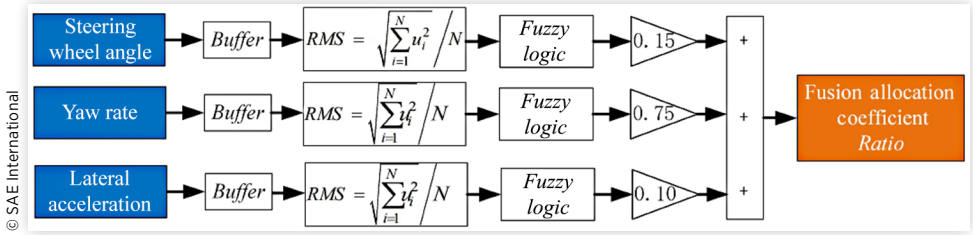
$$V_y_\text{Fusion} = Ratio * V_y_\text{IFC} + \left(1 - Ratio\right) * V_y_\text{Kinematic} \quad \text{Eq. (20)}$$

Tire Parameter Online Adaptive Module

Cornering stiffnesses of front and rear axles are the important parameters of the vehicle dynamic model, which represent the proportional relationship between sideslip angle and tire lateral force. In order to prevent the influence of the change of cornering stiffness on the accuracy of the vehicle dynamic

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FIGURE 5 Calculation of fusion allocation coefficient *Ratio* for lateral velocity fusion estimation.



model, TPOAM is proposed in this article. The TPOAM designed in this article is also an observer-based method, which uses DEKF to identify both vehicle state and tire cornering stiffnesses. At each time step, the state KF uses the current estimated cornering stiffnesses to estimate the state, while the parameter KF uses the currently estimated state to adapt the cornering stiffnesses to reduce error between the estimated state and the measured state. The adaptation steps of TPOAM are shown in Figure 6. The online adaptation of cornering stiffness in TPOAM can be divided into four steps: state prediction, state correction, parameter prediction, and parameter correction.

The state prediction equation is shown as follows:

$$\boldsymbol{x}_{k+1} = \boldsymbol{A}_k \boldsymbol{x}_k + \boldsymbol{B}_k \boldsymbol{u}_k$$
$$\boldsymbol{x}_k = \left[V_y, r\right], \quad \boldsymbol{u}_k = \delta_f$$
$$\boldsymbol{A}_k = \begin{bmatrix} 1 + T_{IMU}\dfrac{C_f + C_r}{mV_x} & T_{IMU}\left(\dfrac{l_f C_f - l_r C_r}{mV_x} - V_x\right) \\ T_{IMU}\dfrac{l_f C_f - l_r C_r}{I_z V_x} & 1 + T_{IMU}\dfrac{l_f^2 C_f + l_r^2 C_r}{I_z V_x} \end{bmatrix} \quad \text{Eq. (21)}$$
$$\boldsymbol{B}_k = \begin{bmatrix} -T_{IMU}\dfrac{C_f}{m} \\ -T_{IMU}\dfrac{l_f C_f}{I_z} \end{bmatrix}$$
$$\boldsymbol{y}_k = \boldsymbol{C}\boldsymbol{x}_k, \quad \boldsymbol{y}_k = \left[V_y_\text{Fusion}, r\right], \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

where δ_f represents the steering wheel angles of the front axles; m, l_f, l_r, I_z represent the vehicle total mass, distance from the C.G. to the front/rear axle, and vehicle inertia moment around z axis, respectively; $W_2 = [W_\beta, W_{V_x}]^T$ is the process noise; $V_2 = [V_{a_x}, V_{V_x}]^T$ is the measurement noise; W_2, V_2 are regarded as white noise with covariance Q_2, R_2, respectively.

State prediction:

$$\hat{\boldsymbol{x}}_k^- = \boldsymbol{A}_{k-1}\hat{\boldsymbol{x}}_{k-1} + \boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1}$$
$$\boldsymbol{P}_{x_k}^- = \boldsymbol{A}_{k-1}\boldsymbol{P}_{x_{k-1}}\boldsymbol{A}_{k-1}^T + \boldsymbol{R}_s^{sys} \quad \text{Eq. (22)}$$

where $\hat{\boldsymbol{x}}_k^-, \hat{\boldsymbol{x}}_k$ represent a priori estimate and a posteriori estimate of the states, respectively; $\boldsymbol{P}_{x_k}^-, \boldsymbol{P}_{x_k}$ represent a priori covariance and a posteriori covariance of the states, respectively; $\boldsymbol{R}_s^{sys}$ represent the system noise covariance of states; $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{x}_k$, $\boldsymbol{u}_k$ is shown in Equation 21.

State correction:

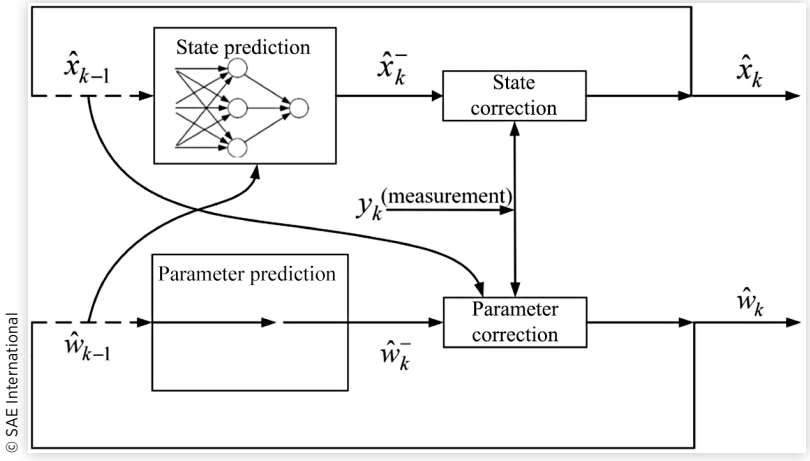
$$\boldsymbol{K}_k^x = \boldsymbol{P}_{x_k}^- \boldsymbol{C}^T \left(\boldsymbol{C}\boldsymbol{P}_{x_k}^- \boldsymbol{C}^T + \boldsymbol{R}_s^{mea}\right)^{-1}$$
$$\hat{\boldsymbol{x}}_k = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k^x \left(\boldsymbol{y}_k - \boldsymbol{C}\hat{\boldsymbol{x}}_k^-\right) \quad \text{Eq. (23)}$$
$$\boldsymbol{P}_{x_k} = \left(\boldsymbol{I} - \boldsymbol{K}_k^x \boldsymbol{C}\right)\boldsymbol{P}_{x_k}^-$$

where $\boldsymbol{K}_k^x$ represent the Kalman gain for states estimation; $\boldsymbol{R}_s^{mea}$ represent the measurement noise covariance of states.

Parameter prediction:

$$\hat{\boldsymbol{w}}_k^- = \hat{\boldsymbol{w}}_{k-1}$$
$$\boldsymbol{P}_{w_k}^- = \boldsymbol{P}_{w_{k-1}} + \boldsymbol{R}_w^{sys} \quad \text{Eq. (24)}$$

FIGURE 6 Online adaptation steps of the TPOAM.



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where $\hat{\boldsymbol{w}}_k^-, \hat{\boldsymbol{w}}_k$ represent a priori estimate and a posteriori estimate of the parameters, respectively; $\boldsymbol{P}_{w_k}^-, \boldsymbol{P}_{w_k}$ represent a priori covariance and a posteriori covariance of the parameters, respectively; $\boldsymbol{R}_w^{sys}$ represent the system noise covariance of parameters. In the cornering stiffness adaptation, $\boldsymbol{w}_k = [C_f; C_r]$. Because the cornering stiffnesses of the front and rear axles change randomly in the process of lateral motion control, the change of the cornering stiffnesses of the front and rear axles are modeled as a random walk behavior, and the magnitude of $\boldsymbol{R}_w^{sys}$ represents the change speed of the cornering stiffnesses of the front and rear axles.

Parameter correction:

$$
\begin{aligned}
\boldsymbol{C}_k^w &= \boldsymbol{C}\frac{\partial \hat{\boldsymbol{x}}_k^-}{\partial \boldsymbol{w}}\bigg|_{\boldsymbol{w}=\hat{\boldsymbol{w}}_k^-} \\
&= \begin{bmatrix} \left(\frac{T_{IMU}l_f l_r}{I_z} - \frac{T_{IMU}}{m}\right)\left(\delta_f - \frac{\hat{V}_{y,k}^- + l_f \hat{r}_k^-}{V_x}\right) & \left(\frac{T_{IMU}}{m} + \frac{T_{IMU}l_r^2}{I_z}\right)\frac{\left(\hat{V}_{y,k}^- - l_r \hat{r}_k^-\right)}{V_x} \\ -\frac{T_{IMU}l_f}{I_z}\left(\delta_f - \frac{\hat{V}_{y,k}^- + l_f \hat{r}_k^-}{V_x}\right) & -\frac{T_{IMU}l_r\left(\hat{V}_{y,k}^- - l_f \hat{r}_k^-\right)}{I_z V_x} \end{bmatrix} \\
\boldsymbol{K}_k^w &= \boldsymbol{P}_{w_k}^-\left(\boldsymbol{C}_k^w\right)^T\left[\boldsymbol{C}_k^w \boldsymbol{P}_{w_k}^-\left(\boldsymbol{C}_k^w\right)^T + \boldsymbol{R}_w^{mea}\right]^{-1} \\
\hat{\boldsymbol{w}}_k &= \hat{\boldsymbol{w}}_k^- + \boldsymbol{K}_k^w \cdot \left(\boldsymbol{y}_k - \boldsymbol{C}\hat{\boldsymbol{x}}_k^-\right) \\
\boldsymbol{P}_{w_k} &= \left(\boldsymbol{I} - \boldsymbol{K}_k^w \boldsymbol{C}\right)\boldsymbol{P}_{w_k}^-
\end{aligned}
\qquad \text{Eq. (25)}
$$

where $\boldsymbol{K}_k^w$ represents the Kalman gain for parameters estimation; $\boldsymbol{R}_w^{mea}$ represents the measurement noise covariance of parameters. Since the measurement data of state KF and parameter KF are both the lateral velocity and yaw rate at the rear axle, the measurement noise covariance $\boldsymbol{R}_s^{mea}, \boldsymbol{R}_w^{mea}$ are the same which is mainly determined by the measurement accuracy of the vehicle status sensor. The system noise covariance $\boldsymbol{R}_s^{sys}, \boldsymbol{R}_w^{sys}$ are mainly determined by the errors of the system state space equations in the state KF and parameter KF; $\boldsymbol{R}_s^{sys}$ represents the prediction accuracy of the single-track dynamic model for the lateral velocity and yaw rate at the vehicle center of mass. The greater it is relative to the measurement noise, the worse the prediction accuracy of the single-track dynamic model. Since it is assumed that the change of parameters $\boldsymbol{w}_k$ in the lateral motion control process is a random walk process; $\boldsymbol{R}_w^{sys}$ mainly represents the intensity of the change of the parameters. The larger $\boldsymbol{R}_w^{sys}$, the stronger the parameters $\boldsymbol{w}_k$ change.

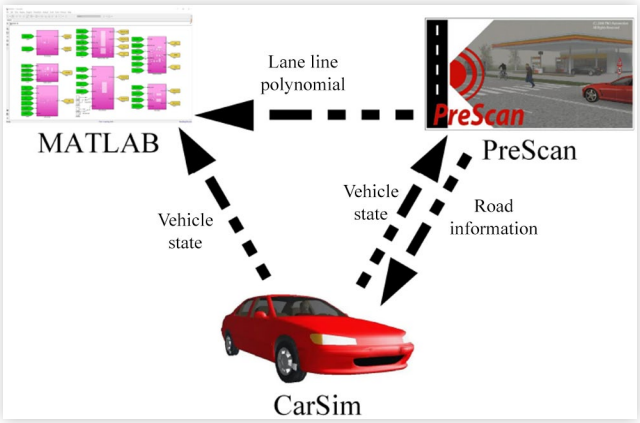
In the process of tire parameter online adaptation, it should be noted that the cornering stiffnesses cannot be accurately estimated under all working conditions, that is, the cornering stiffnesses are not observable under all working conditions. When the vehicle travels along a straight line, the front wheel angle δ_f, lateral speed V_y, and yaw rate r all tend to zero. As can be seen from Equation 25, at this time, $\boldsymbol{C}_k^w$ tends to zero matrices, and the adaptation of cornering stiffnesses will fail. To address the above issue, the norm of $\boldsymbol{C}_k^w$ is calculated in this article. When the norm of $\boldsymbol{C}_k^w$ exceeds the threshold value $NORM_{thre}$, which means the vehicle has sufficient lateral excitation, TPOAM is activated. When the norm of $\boldsymbol{C}_k^w$ is less than the threshold value $NORM_{thre}$, it indicates that there is no sufficient lateral excitation. At this time, TPOAM will directly output the nominal tire cornering stiffnesses $\boldsymbol{w}_{nominal}$ to avoid the wrong estimation of tire cornering stiffnesses. At the same time, to prevent the sudden change of cornering stiffness switching from affecting the lateral motion control, the first-order inertia filter is used to filter it to get to the final cornering stiffnesses adaptation result $\boldsymbol{w}_{final}$, as shown in Equation 26, where τ_f is the time constant of the first-order inertia filter, which is taken as 0.2 s in this article.

$$
\begin{aligned}
\boldsymbol{w}_{output} &= \begin{cases} \hat{\boldsymbol{w}}_k, & norm\left(\boldsymbol{C}_k^w\right) > NORM_{thre} \\ \boldsymbol{w}_{nominal}, & norm\left(\boldsymbol{C}_k^w\right) \le NORM_{thre} \end{cases} \\
\boldsymbol{w}_{final} &= \frac{1}{\tau_f s + 1}\boldsymbol{w}_{output}
\end{aligned}
\qquad \text{Eq. (26)}
$$

Simulation and Results

In order to validate the estimation algorithm of key parameters in vehicle dynamics based on multi-source information fusion designed in this article, the PreScan-CarSim-MATLAB co-simulation platform is built in this article, as shown in Figure 7. MATLAB integrates the co-simulation environment and implements the estimation algorithm designed in this article. PreScan is used to set up the road environment and outputs the detected lane line polynomial coefficient information to MATLAB, then needs to output the relevant information of the road environment to the CarSim. CarSim provides the vehicle dynamics model and outputs the measurable vehicle state to the MATLAB and PreScan software. The vehicle parameters are shown in Table 1. In order to simulate the real IMU measurement characteristics, white noise is injected into the longitudinal acceleration, lateral acceleration,

FIGURE 7 PreScan-CarSim-MATLAB co-simulation platform.



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TABLE 1 Parameters of the tested vehicle.

Parameters	Symbol	Unit	Value
Mass of the vehicle	m	kg	2020
Wheelbase of the vehicle	L	m	2.947
Distance from the C.G. to the front axle	l_f	m	1.265
Distance from the C.G. to the rear axle	l_r	m	1.682
Vehicle moment of inertia around z axis	I_z	kg*m^2	4095
Update cycle of IFC sensor	T_{IFC}	s	0.05
Update cycle of IMU sensor	T_{IMU}	s	0.01

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and yaw rate outputted by CarSim to simulate the real IMU measurement characteristics. The injected noise information is shown in Table 2.

Double Lane Change Simulation Test

In this section, the proposed algorithm is validated in the DLC condition, and the target path is set with reference to the DLC condition in ISO 3888. Under the simulation condition, the vehicle's longitudinal velocity is set as 90 km/h. The coefficient of road surface adhesion is set as constant $\mu = 0.85$. The road width is set at a standard lane width of 3.75 m. The simulation results are shown in Figures 8 and 9.

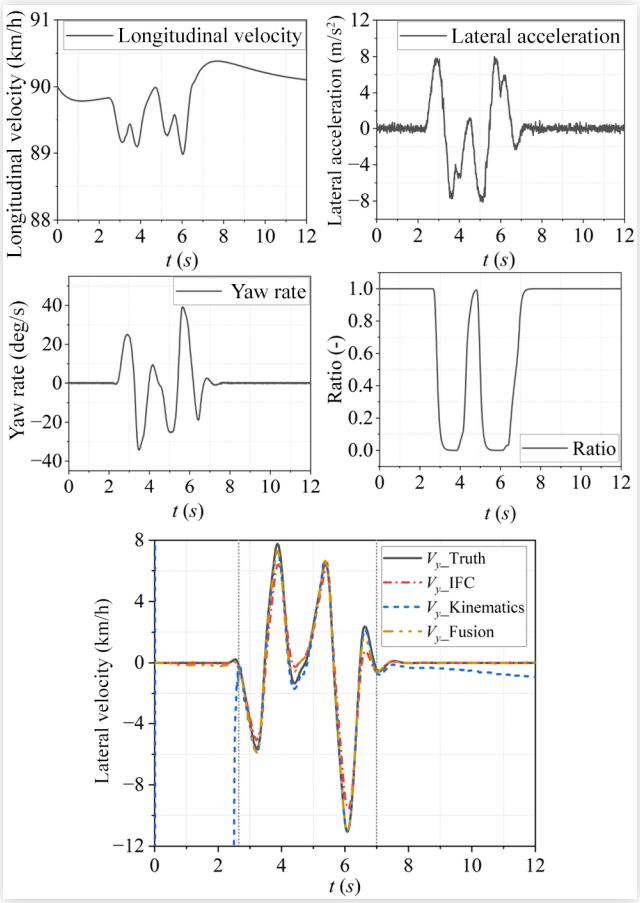
As shown in Figure 8, the vehicle longitudinal velocity remains constant at 90 km/h throughout the whole DLC condition. The vehicle's lateral acceleration absolute value reaches a maximum of 8.1 m/s^2. The results of the vehicle lateral velocity estimation based on the kinematic model, with the vehicle state input containing white noise, undergo a severe drift in the 0–2.7 s after the start of the simulation, as shown in Figure 8. Meanwhile, in the 7–12 s time after the start of the simulation, the vehicle lateral velocity estimation based on the kinematic model is also shifted compared to the actual lateral velocity and cannot converge. In the 7–12 s time after the start of the simulation, the vehicle lateral velocity and the sideslip angle change drastically. At this time, the results of the vehicle lateral velocity estimation based on the kinematic model can be quickly and accurately converged to the truth value of the vehicle lateral velocity. Vehicle lateral velocity estimation based on IFC sensor accurately tracks the

TABLE 2 The injected noise information.

Signal	Upper limit of noise	Lower limit of noise	Mean error
Longitudinal acceleration a_x (m/s^2)	1.0	−1.0	0.2
Lateral acceleration a_y (m/s^2)	0.6	−0.6	0.13
Yaw rate r (deg/s)	0.5	−0.5	0.09

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FIGURE 8 Lateral estimation results of the DLC simulation test.



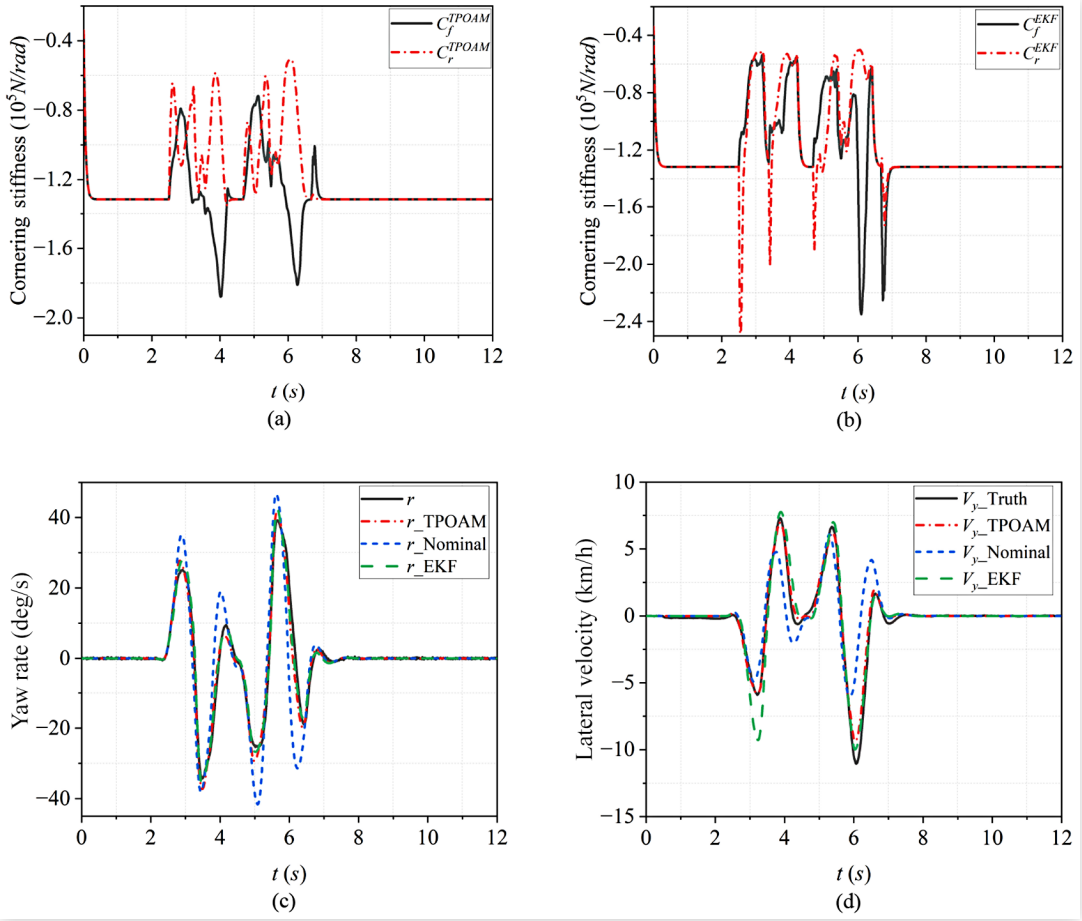
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true lateral velocity throughout the DLC condition. However, under the working conditions where the vehicle lateral speed changes drastically, the accuracy of the lateral velocity estimation results based on IFC sensor is poorer compared to the vehicle lateral speed estimation results based on the kinematic model, for example, at the 3.9 s, 4.4 s, 6.1 s, and 6.65 s after the beginning of the simulation, the vehicle lateral speed estimation results based on IFC sensor have a significant error with the actual vehicle lateral speed, and the absolute error values are, respectively, 1.4 km/h, 1.1 km/h, 1.5 km/h, and 1.7 km/h.

The vehicle lateral velocity estimation results under multi-source information fusion, and the fusion allocation coefficient *Ratio* of the vehicle lateral velocity estimation results are shown in Figure 8. According to the vehicle lateral dynamic characteristics, the fusion allocation coefficient *Ratio* tends to be 1 in the vehicle steady-state condition (within the 0–2.7 s and the 7–12 s after the simulation start), which means the fusion estimation results adopt the vehicle lateral velocity estimation results based on IFC sensor. And when the vehicle lateral velocity changes drastically during the DLC simulation (within the 2.7–7 s after the simulation start), the fusion coefficient *Ratio* tends to be 0. At this time, the fusion estimation results adopt the vehicle lateral velocity estimation results

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FIGURE 9 Verification of the estimated cornering stiffnesses for DLC condition: (a) cornering stiffness output by TPOAM; (b) cornering stiffness output by EKF; (c) predicted yaw rate; (d) predicted lateral velocity.



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based on the kinematic model. The RMSE of the vehicle lateral speed estimation results V_y_IFC, V_y_Kinematic, V_y_Fusion are 0.5079 km/h, 309.2221 km/h, and 0.2114 km/h, respectively, as shown in Table 3. Through the multi-source information fusion algorithm, the average error of the lateral speed estimation results can be effectively suppressed, achieving accurate estimation of vehicle lateral velocity.

To verify the effectiveness of the cornering stiffnesses output from the TPOAM designed in this article, we utilize the lateral acceleration and yaw rate of the vehicle output from the IMU as inputs, and estimate the cornering stiffnesses of the front and rear axles using the EKF without vehicle lateral velocity measurement, and the outputs are denoted as C_f^{EKF},C_r^{EKF}. Since the actual values of the equivalent lateral stiffness of the front and rear axles cannot be directly obtained, in order to verify the tire parameter online adaptation results of TPOAM proposed in this article, the three-degree-of-freedom vehicle dynamic model is adopted as the reference model. The steering angles of the front and rear axles and the four in-wheel motor's driving torques are used as known inputs to calculate the vehicle state simultaneously. The calculated vehicle states are compared with the actual vehicle states output from the CarSim and the comparison results are shown in Figure 9.

TABLE 3 The RMSE of estimation results under DLC test.

	V_y_IFC	V_y_Kinematics	V_y_Fusion
Lateral velocity estimation RMSE (km/h)	0.5079	309.2221	**0.2114**
	TPOAM	Nominal	EKF
Yaw rate prediction RMSE (deg/s)	1.8059	6.2648	**1.2961**
Lateral velocity prediction RMSE (km/h)	**0.4111**	1.6558	0.6759

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Within the time range of 0–2.7 s and 7–12 s after the beginning of the simulation, using the nominal cornering stiffnesses and the cornering stiffnesses obtained from TPOAM as parameters, the lateral velocity and the yaw rate calculated by the vehicle dynamic model almost coincide, and the difference with the actual vehicle states output by CarSim is small. This is because there is no sufficient lateral excitation in the above two time ranges, and the outputs of TPOAM are just the nominal cornering stiffnesses. The nominal cornering

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stiffnesses can accurately reflect the proportional relationship between the tire lateral force and the tire sideslip angle in the case of a small sideslip angle. The vehicle has significant lateral velocity and yaw rate during the 7–12 s after the start of the simulation. At this time, TPOAM outputs the results of cornering stiffnesses obtained by DEKF. Whether in the steady state or the transient change state, using the cornering stiffness obtained by TPOAM as the parameters, the lateral velocity and yaw rate calculated by the three-degree-of-freedom vehicle dynamic model are closer to the actual vehicle state than the calculated results taking the nominal cornering stiffnesses as the parameters, as shown in Figure 9(c) and (d). The RMSE for lateral velocity and yaw rate, when using the nominal value of cornering stiffnesses, is 1.6558 km/h and 6.2648 deg/s, respectively. In contrast, when using cornering stiffnesses determined by the TPOAM, the RMSE is significantly reduced to 0.4111 km/h for lateral velocity and 1.8059 deg/s for yaw rate. On the other hand, employing the cornering stiffnesses from the EKF results in an RMSE of 0.6759 km/h for lateral velocity and 1.2961 deg/s for yaw rate. When compared to the TPOAM, the predictions from the EKF exhibit a larger error in lateral velocities. This discrepancy can be attributed to the EKF not having an accurate lateral velocity input. In summary, the TPOAM, as presented in this study, can adeptly and promptly adjust the tire cornering stiffnesses, leading to a marked improvement in the precision of the vehicle dynamic model.

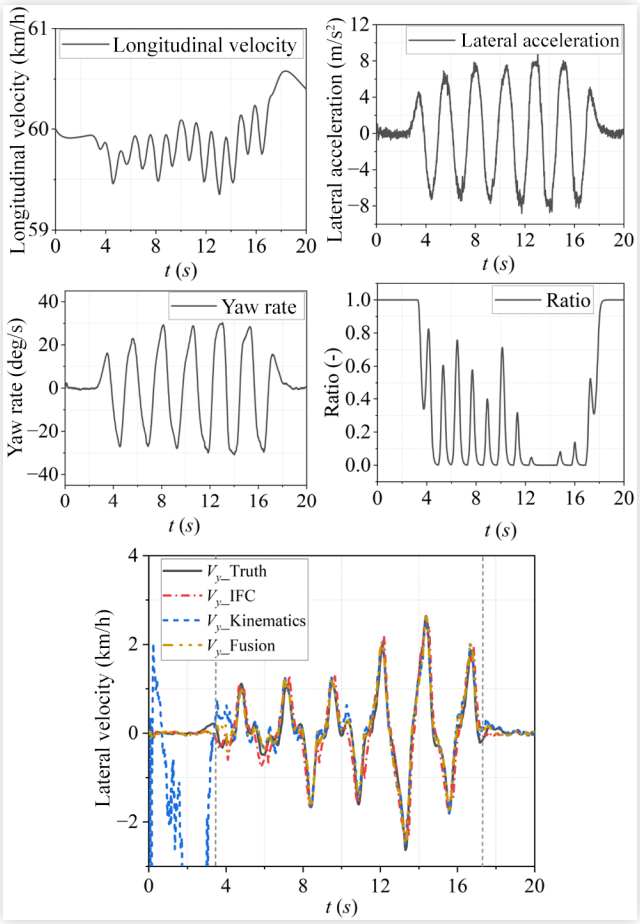
Slalom Simulation Test

In this section, the proposed algorithm is validated in the slalom condition, and the vehicle's longitudinal velocity is set as 60 km/h. The coefficient of road surface adhesion is set as constant μ = 0.85. The road width is set at a standard lane width of 3.75 m. The simulation results are shown in Figures 10 and 11.

As shown in Figure 10, the vehicle longitudinal velocity remains constant at 60 km/h throughout the whole slalom condition. The vehicle's lateral acceleration reaches an absolute maximum of 8.7 m/s^2. The results of the vehicle lateral velocity estimation based on the kinematic model, with the vehicle state input containing white noise, undergo a severe drift in the 0–4.5 s after the start of the simulation, as shown in Figure 10. In the 4.5–17.3 s time after the start of the simulation, the vehicle lateral velocity and the sideslip angle change drastically. At this time, the results of the vehicle lateral velocity estimation based on the kinematic model can be quickly and accurately converged to the truth value of the vehicle lateral velocity.

Vehicle lateral velocity estimation based on IFC sensor accurately tracks the true lateral velocity throughout the slalom condition. However, under the working conditions where the vehicle lateral speed changes drastically, the accuracy of the lateral speed estimation results based on IFC sensor is poorer compared to the vehicle lateral speed

FIGURE 10 Lateral estimation results of the slalom simulation test.



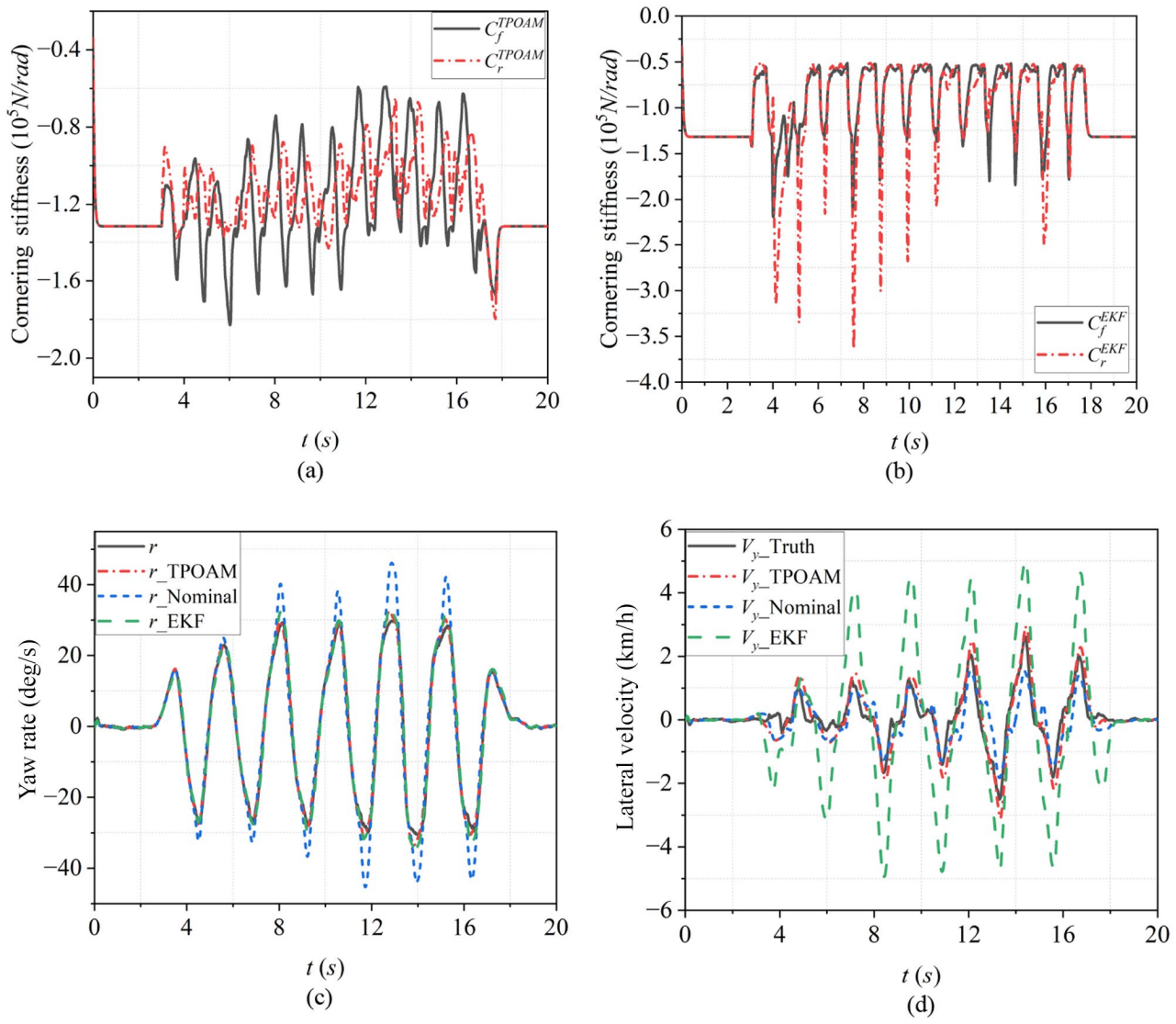
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estimation results based on the kinematic model, for example, at the 11.2 s and 13.5 s, after the beginning of the simulation, the vehicle lateral speed estimation results based on IFC sensor have a significant error with the actual vehicle lateral speed, and the absolute error values are, respectively, 0.85 km/h and 0.91 km/h.

The vehicle lateral velocity estimation results under multi-source information fusion, and the fusion allocation coefficient *Ratio* of the vehicle lateral velocity estimation results are shown in Figure 10. According to the vehicle lateral dynamic characteristics, the fusion allocation coefficient *Ratio* tends to be 1 in the vehicle steady-state condition (within the 0–3.5 s and the 18–20 s after the simulation start), which means the fusion estimation results adopt the vehicle lateral velocity estimation results based on IFC sensor. And when the vehicle lateral velocity changes drastically during the slalom simulation (within the 3.5–18 s after the simulation start), the fusion coefficient *Ratio* tends to be 0. At this time, the fusion estimation results adopt the vehicle lateral velocity estimation results based on the kinematic model. The RMSE of the vehicle lateral speed estimation results V_{y}_IFC, V_{y}_Kinematic, V_{y}_Fusion are 0.2451 km/h, 312.2250 km/h, and 0.1400 km/h, respectively, as shown in Table 4. Through the

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FIGURE 11 Verification of the estimated cornering stiffnesses for slalom condition: (a) cornering stiffness output by TPOAM; (b) cornering stiffness output by EKF; (c) yaw rate; (d) lateral velocity.



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multi-source information fusion algorithm, the average error of the lateral speed estimation results can be effectively suppressed.

The comparison results to verify the effectiveness of the proposed TPOAM algorithm in this article are shown in Figure 11. Within the time range of 0–3.1 s and 18–20 s after the beginning of the simulation, using the nominal cornering stiffnesses and the cornering stiffnesses obtained from TPOAM as parameters, the lateral velocity and the yaw rate calculated by the vehicle dynamic model almost coincide, and the difference with the actual vehicle states output is small. There is no sufficient lateral excitation in the above two time ranges, and the outputs of TPOAM are just the nominal cornering stiffnesses. The vehicle has significant lateral velocity and yaw rate during the 3.1–18 s after the start of the simulation. TPOAM outputs the results of cornering stiffnesses obtained by DEKF. Whether in the steady state or the transient change state, using the cornering stiffness obtained by TPOAM as the parameters, the lateral velocity and yaw rate calculated by the three-degree-of-freedom vehicle dynamic model are closer to the actual vehicle state than the

TABLE 4 The RMSE of estimation results under slalom test.

	V_y_IFC	V_y_Kinematics	V_y_Fusion
Lateral velocity estimation RMSE (km/h)	0.2451	312.2250	**0.1400**
	TPOAM	Nominal	EKF
Yaw rate prediction RMSE (deg/s)	**1.0124**	4.7246	1.6609
Lateral velocity prediction RMSE (km/h)	**0.31083**	0.44583	1.5817

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calculated results taking the nominal cornering stiffnesses as the parameters. The RMSE of lateral velocity and yaw rate calculated by using the nominal value of cornering stiffnesses as a parameter are 0.4458 km/h and 4.7246 deg/s, respectively. In comparison, the RMSE of lateral velocity and yaw rate calculated by using the cornering stiffnesses obtained by the TPOAM as a parameter is 0.3108 km/h and 1.0124 deg/s, respectively. The RMSE for lateral velocity and yaw rate, when derived using cornering stiffnesses determined by the EKF, is 1.5817 km/h and 1.6609 deg/s, respectively. In comparison with TPOAM, the predictions for both lateral velocity and yaw rate from the EKF show significant errors, particularly during abrupt variations in the vehicle's lateral velocity. This discrepancy arises because the EKF lacks precise lateral velocity input. The TPOAM designed in this article can accurately and in real-time adapt the tire cornering stiffnesses and effectively improve the accuracy of the vehicle dynamic model in the slalom condition.

Conclusions

In this article, a multi-source information fusion method is designed for the estimation of key parameters of vehicle dynamics. First, by utilizing the lane line polynomial information obtained from the detection of the IFC sensor in the autonomous driving domain, the observation of the vehicle's lateral speed is achieved without any additional sensor cost. Second, the kinematic-model-based estimation approach uses the vehicle's measurable lateral acceleration, longitudinal velocity, longitudinal acceleration, and yaw rate as inputs to estimate the vehicle's sideslip angle as well as lateral velocity. By comparing the two lateral velocity estimation methods, this article designs the vehicle lateral velocity estimation method based on multi-source information fusion. By extracting the lateral dynamic characteristics of the vehicle and calculating the fusion allocation coefficient *Ratio*, the results of the two lateral velocity estimation methods are fused to realize the rapid convergence of the lateral velocity estimation in the steady-state condition and the accurate tracking of the lateral velocity in the dynamic condition. In this article, TPOAM is also designed to update the dynamics key parameter cornering stiffness online. Finally, the proposed algorithm is verified in the DLC and slalom simulation tests.

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